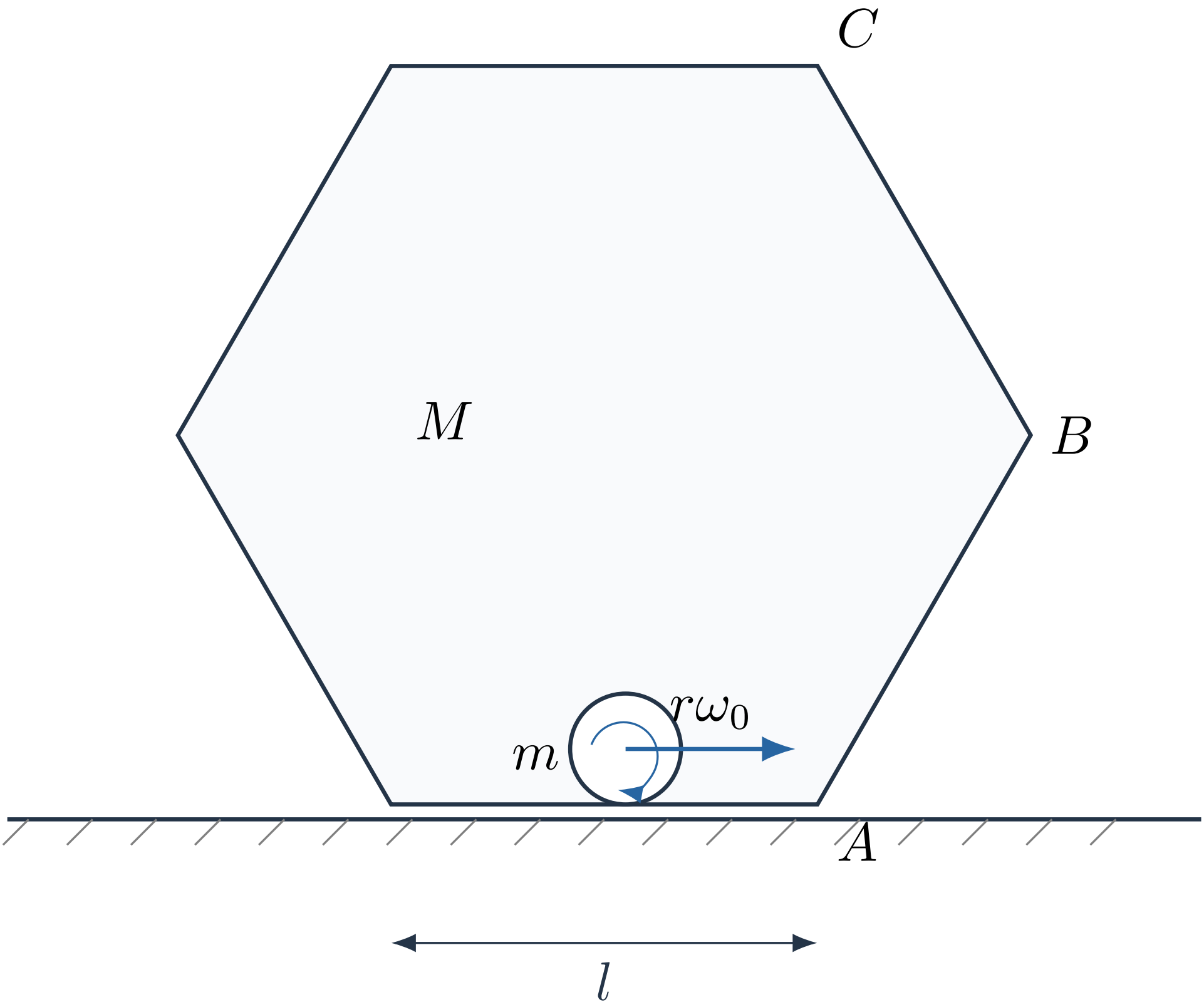
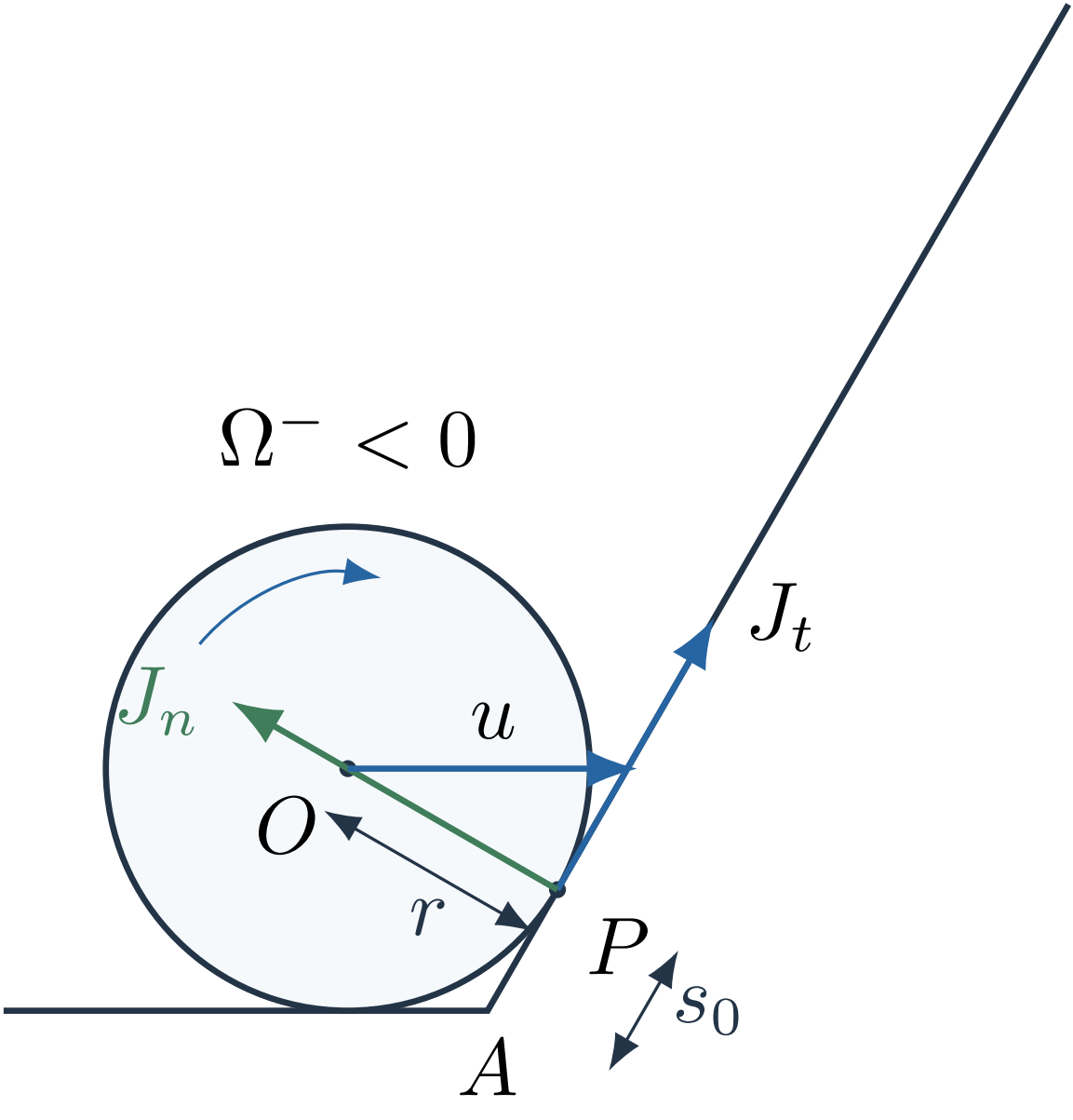
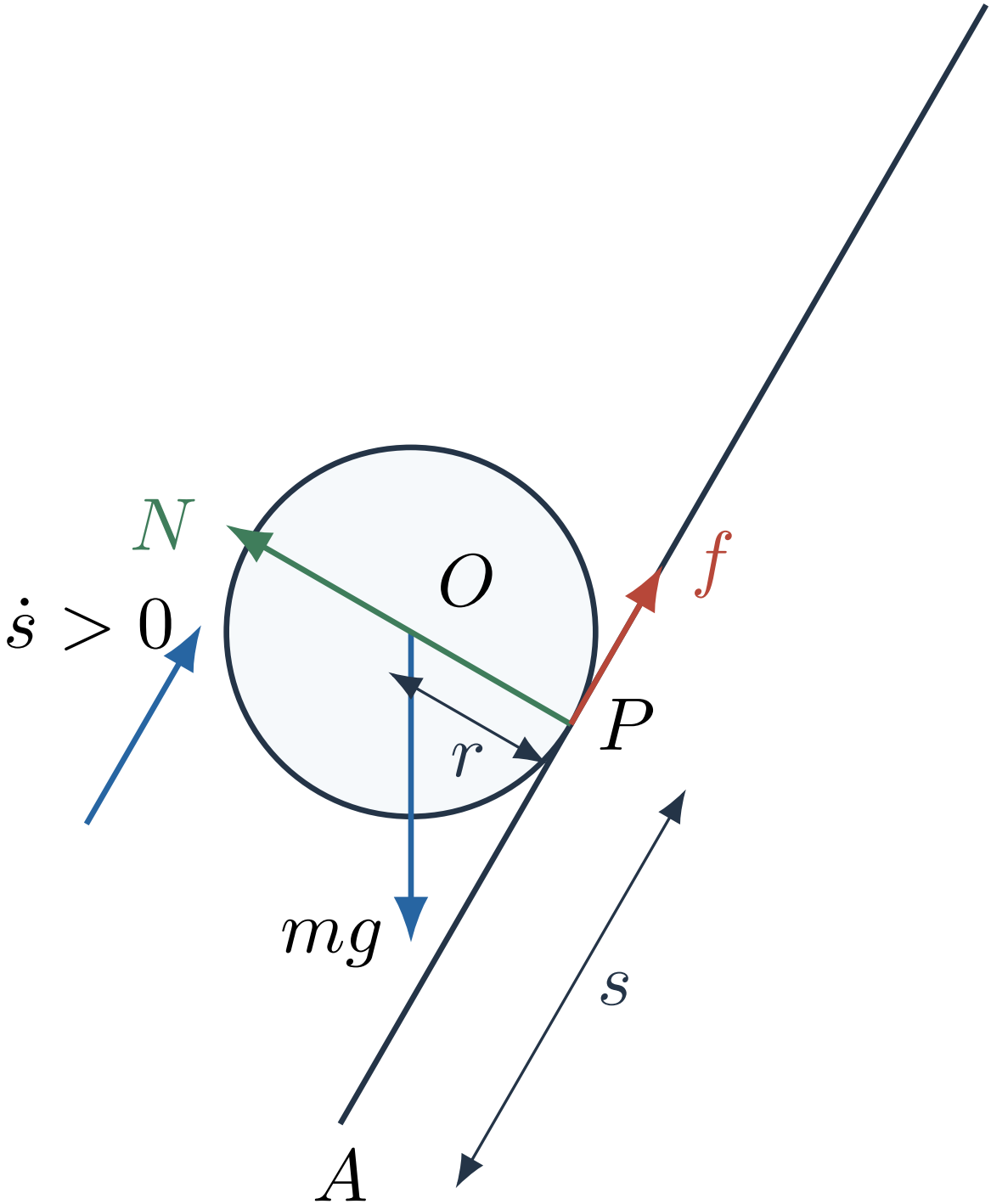
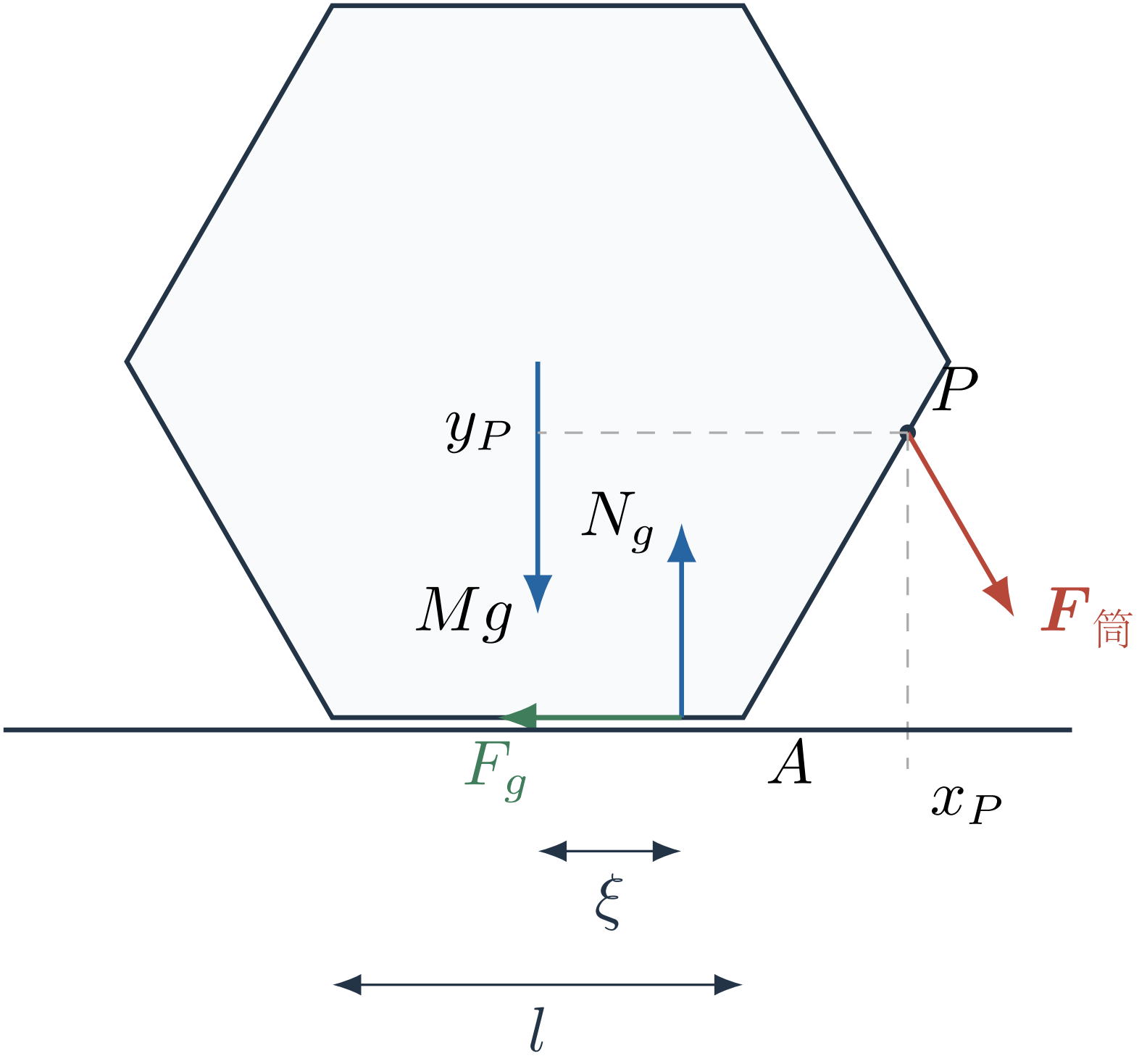

C
M
B
$r\omega_0$
m
A
l


$\Omega^- < 0$
J_t
J_n
u
O
r
P
s_0
A


N
O
f
$\dot{s} > 0$
P
r
mg
s
A


P
y_P
N_g
Mg
$\boldsymbol{F}$筒
F_g
A
x_P
ξ
l